

ALESSIA VISCONTI, PhD

Assistant professor (tenure track)

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RESEARCH INTERESTS

- BIOSTATISTICS & BIOINFORMATICS
- COMPUTATIONAL MEDICINE
- DATA MINING & MACHINE LEARNING
- MULTI-OMICS DATA ANALYSIS AND INTEGRATION
- RESEARCH SOFTWARE ENGINEERING

BRIEF DESCRIPTION

Alessia Visconti is an expert in biostatistics and computational medicine. Her research deals with the development and application of statistical and computational methods to identify multi-*omics* modifications influencing complex human phenotypes. She has also worked on the problem of knowledge discovery in biological data, developing new approaches to solve biological tasks, and on the reverse engineering of gene regulatory networks.

EDUCATION

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| JUL 2012 | PhD in Science and High Technology (Specialisation: Computer Science)
Department of Computer Science, University of Turin |
| JUL 2008 | MSc in Computer Science “ <i>summa cum laude</i> ”
Department of Computer Science, University of Turin |
| MAR 2006 | BSc in Computer Science “ <i>summa cum laude</i> ”
Department of Computer Science, University of Turin |

SKILLS

LANGUAGE SKILLS	ITALIAN: native proficiency ENGLISH: professional working proficiency GREEK (MODERN): elementary proficiency
COMPUTING SKILLS	PROGRAMMING LANGUAGES: bash, C, C++, JAVA, python, R, ruby OTHER LANGUAGES: CSS, L ^A T _E X, HTML STATISTICAL SOFTWARE: R DATABASE MANAGEMENT: SQL VERSION CONTROL & REPRODUCIBILITY : GIT, nextflow, docker, singularity BIOINFORMATICS & GENETIC ANALYSIS: BBmap, BEDTools, DESeq2, GCTA, GEMMA, GWAMA, LDAK, limma, lme4, metal, Merlin, PLINK, QTD, samtools, vcftools, ... STRUCTURAL EQUATION MODELLING: openMX DATA VISUALISATION: dot, ggplot2

- FEB 2025 - PRESENT** **Visiting Senior Lecturer** at the Department of Twin Research & Genetic Epidemiology, King's College London
RESEARCH ACTIVITY: Alessia Visconti continues her research in (a) IgA glycosylation [J56], (b) genetic epidemiology [J54, J58, J63], and (d) the gut microbiome and its effects on human health [J52, J53, J55, J57, J61, J64], with a particular interest in its phage component [J51, PP1].
NOTE: Alessia Visconti was *Visiting Research Fellow* at the same institution from October 2023 to September 2024.
- DEC 2023 - PRESENT** **Assistant professor (tenure track)** at the Center for Biostatistics, Epidemiology and Public Health, Department of Clinical and Biological Sciences, University of Turin
RESEARCH ACTIVITY: Alessia Visconti's research focuses on (a) the development and application of statistical and computational multi-omics approaches to understand the underlying mechanisms, and to improve the diagnosis and treatment of complex human diseases, (b) the design and deployment of deep learning approaches for monitoring and improving public health responses [J59, J62, PP2, P6], and (c) the integration of patient preferences into clinical care and drugs and medical devices development [J60]. She continues to teach undergraduate and postgraduate courses, supervise PhD students and postdoctoral researchers, and serve as MSc thesis supervisor. She contributes to the design and writing of international funding proposals, for which she has scientific responsibility.
- OCT 2023 - NOV 2023** **Senior bioinformatician** at the Genomics Research Centre, Human Technopole, Milan
RESEARCH ACTIVITY: Alessia Visconti was involved in a project conducting a genome-wide association study (GWAS) on phenotypes extracted from cardiac MRI scans using an unsupervised deep learning approach [J65].
- AUG 2022 - APR 2023** **Career break:** Maternity leave
- AUG 2017 - SEP 2023** **Research fellow** at the Department of Twin Research & Genetic Epidemiology, King's College London
RESEARCH ACTIVITY: Alessia Visconti's research focused on the development and application of statistical and computational multi-omics approaches to understand the underlying mechanisms, and to improve the diagnosis and treatment of complex human diseases. She continued to teach postgraduate courses, supervise PhD students and postdoctoral researchers and serve as MSc thesis supervisor. She contributed to the design and writing of international funding proposals, for which she had scientific responsibility.
 Building on her previous position, she collaborated on projects studying IgA nephropathy [J37] as well as melanoma and its risk factors, particularly from a genomic perspective [J20, J23, J24, J28], and begun research on predicting the response to immunotherapy in melanoma patients [J42, J45]. She also studied (a) the influence of the gut microbiome on human health [J21, J30, J40, J43, J44, J47, J48, J49], (b) the multi-omics profiles underlying chronic autoimmune thyroiditis [J22], (c) the genomic basis of atopic dermatitis [J38, J50], (d) the influence of amylase copy-number variation (CNV) on a range of adiposity traits [J34], (e) the effect of cigarette smoking on more than 35,000 immunophenotypes [J35], and (f) the effect of X-chromosome inactivation [J46]. During the SARS-CoV-2 pandemic, she collaborated within the ZOE COVID Study (now ZOE Health Study) to develop the pipeline for analysing data submitted daily by approximately 4 million users [J39]. She also led two studies on the cutaneous symptoms of the infection [J32, J41], and collaborated as data analyst in several other studies [J25, J27, J29, J31, J33, J36].
- APR 2015 - JUL 2017** **Research associate** at the Department of Twin Research & Genetic Epidemiology, King's College London

RESEARCH ACTIVITY: Alessia Visconti’s research focused on the development and application of statistical and computational multi-*omics* approaches to understand the underlying mechanisms, and to improve the diagnosis and treatment of complex human diseases. In particular, she led the statistical analysis in a series of projects aimed at studying IgA Nephropathy [J9], cognition and neurodevelopmental disorders [J19], and epigenetic modification [J13] as well as at dissecting the aetiology of melanoma, melanoma risk phenotypes, and their connection with ageing [J10, J14, J15, J18]. Becoming interested in the human microbiome, she defined a set of guidelines for conducting metagenomic studies in clinical settings [J17], developed a reproducible pipeline for the analysis of metagenomic data (*YAMP*) [J16], and became the scientific lead for the analysis of metagenomic data at the Department of Twin Research & Genetic Epidemiology, overseeing the processing and management of more than 4,000 samples (an activity that is still ongoing as part of Alessia Visconti’s position as Visiting Senior Lecturer in the Department). She began to teach postgraduate courses, supervise PhD students and post-doctoral researchers and serve as MSC thesis supervisors. She contributed to the design and writing of international funding proposals, for which she had scientific responsibility.

- JAN 2014 - MAR 2015 **Research associate** at the Department of Genomics of Common Disease, School of Public Health, Imperial College London
RESEARCH ACTIVITY: Alessia Visconti developed and implemented an approach for the association testing of quantitative traits on related individuals (*PopPanTe*) [J12], which was used in one of the first epigenome-wide association studies (EWAS) in a Middle Eastern population [J7]. She collaborated on studies investigating the effect of CNV on reading abilities [J8], and the gene regulatory networks underlying cognitive abilities associated with neurodevelopmental disorders [J6].
- JAN 2012 - DEC 2013 **Research associate** at the Department of Computer Science, University of Turin
RESEARCH ACTIVITY: Alessia Visconti (*a*) developed and implemented a bi-clustering approach which can leverage *a priori* knowledge to identify clinically relevant groups (*AID-ISA*) [J4], (*b*) contributed to the development of an exact algorithm for answering Maximum a Posteriori queries on tree-structured probabilistic graphical models (*CDoT*) [P5], (*c*) applied machine learning approaches for predicting small peptide lipophilicity [J5], (*d*) conducted the bioinformatics analysis to assess the ability of more than 200 compounds to act as hydrogen-bond donors [J3], and (*e*) developed a tool for searching topologies within chemical molecules, which has been used to support teaching at the Department of Molecular Biotechnology and Health Sciences, University of Turin (*PicNic*).
- JUN 2011 - DEC 2011 **Visiting researcher** at the Center of Biological Sequence Analysis, Technical University of Denmark
RESEARCH ACTIVITY: Alessia Visconti used data generated at the Danish centre to infer causal relationships between transcriptomic and proteomic levels using Granger causality testing [T3]. She developed, together with Ali Altıntas and Chris Workman, an *ensemble* approach combining results from regression trees and support vector machine for regression for the “*DREAM6 – Promoter Activity Prediction Challenge*”, which ranked 8th out of 21 participants.
- JAN 2009 - DEC 2011 **PhD student** at Department of Computer Science, University of Turin
RESEARCH ACTIVITY: Alessia Visconti developed and implemented: (*a*) an algorithm for extracting protein motifs and clustering them into families using a constrained co-clustering approach (also providing a web interface, *MotifsLinker*) [P1], (*b*) a co-clustering algorithm embedding Gene Ontology annotations to provide more meaningful grouping (again accompanied by a web interface, *GOClust*) [BC2, P2], (*c*) an algorithm for clustering Gene Ontology terms according to their function to obtain a more compact and informative ontology (*RGO*) [J1, P3], (*d*) two algorithms for the reverse engineering of gene regulatory networks [P4, J2, T3], and (*e*) contributed to the development of a

modular software architecture for the analysis of metagenomic sequences, taking primary responsibility for the co-clustering module [BC1].

RESEARCH COLLABORATIONS

Only collaborators outside King’s College London (for projects started between 2014 and 2023) and the University of Turin (for projects started before 2014 and after 2023) are reported.

PROJECTS FOR WHICH ALESSIA VISCONTI HAS SCIENTIFIC RESPONSIBILITY

- 2025 - 2026 TITLE: Digital transition and artificial intelligence in the social and healthcare sector in the Province of Cuneo: dissemination, best practices, and possible developments
FUNDER: Cassa di Risparmio di Cuneo Foundation; BUDGET: €58,000
ROLE: co-PI
OBJECTIVES: Analysis of the diffusion and socio-economic impact of digital technologies in the social and healthcare sectors in the province of Cuneo, Italy.
SCIENTIFIC CONTRIBUTION: Alessia Visconti co-led an initial review of the scientific and grey literature, co-designed two surveys, one aimed at citizens and the other at health and social care professionals, and will contribute to their quantitative and qualitative analysis. The project will result in the publication of a monograph as part of the “*Quaderni della Fondazione CRC*” series. As part of this project, she co-supervises a PhD student (Emanuele Koumantakis) and a research assistant (Marco Vitturini).
- 2025 - 2026 TITLE: Read Assessment and Decision Support for ICU Readmission Prediction (READ-ICU)
FUNDER: Horizon Enfield – 2nd Exchange Scheme Open call; BUDGET: €43,200 (mobility allowance for three researchers)
COLLABORATORS: National Technical University of Athens, Greece
ROLE: co-PI
OBJECTIVES: Developing an ensemble approach to predict the risk of ICU readmission.
SCIENTIFIC CONTRIBUTION: Alessia Visconti co-leads the development of an approach based on the *Bayesian Model Averaging Framework* to integrates statistical, classical machine learning, and deep learning models to improve the prediction of ICU readmission risk using publicly ICU datasets [PP2]. The models were identified through a systematic review of the literature, which also included a meta-analysis [J59]. As part of this project, she co-supervises a PhD student (Emanuele Koumantakis).
- 2024 - 2026 TITLE: Exploring the potential role of gut microbiota in improving outcomes of chronic myeloid leukaemia patients
FUNDER: ESH John Goldman Research Award; BUDGET: €80,000
COLLABORATORS: Italian Institute for Genomic Medicine, Italy
ROLE: Work package leader (data analysis)
OBJECTIVES: Analysis of the interactions between the gut microbiome and response to pharmacological treatments in patients with chronic myeloid leukaemia.
SCIENTIFIC CONTRIBUTION: Alessia Visconti used unsupervised machine learning techniques and Bayesian statistics to identify gut microbiome profiles associated with a deep molecular response, supporting the hypothesis that the gut microbiome plays a role in modulating drug efficacy.
- 2022 - 2024 TITLE: A Collaborative Approach to the Borne Uterine Mapping Programme (BUMP) Feasibility Study
FUNDER: BORNE medical research charity; BUDGET: £500,000
COLLABORATORS: Imperial College London, UK, Newcastle University, UK, and University of Turin, Italy
ROLE: Work package leader(data analysis).
OBJECTIVES: Analysis of transcriptional profiles associated with the onset of spontaneous labour through the integration of scRNA-seq, snRNA-seq, and spatial transcriptomics data.
SCIENTIFIC CONTRIBUTION: After conducting quality control and preliminary analyses, Alessia Visconti characterised the transcriptional profiles using statistical and

computational approaches to (a) assess the reproducibility of gene expression networks in scRNA-seq and snRNA-seq experiments, (b) identify differentially abundant cell sub-populations in scRNA-seq and snRNA-seq experiments, and (c) identify differentially expressed genes in spontaneous compared with non-spontaneous labour. The study enabled a systematic assessment of the strengths and limitations of single-cell and single-nucleus technologies, showing how their integration with spatial transcriptomics improves the resolution and interpretation of the cellular and tissue mechanisms underlying labour.

- 2021 - 2024 TITLE: Understanding phenotype and mechanisms of spontaneous preterm birth in sub-Saharan Africa (PRECISE-SPTB)
FUNDER: Medical Research Council (MRC); BUDGET: £458,80
COLLABORATORS: Aga Khan University, Kenya, MRC Unit The Gambia at LSHTM, The Gambia, and Eduardo Mondlane University, Mozambique
ROLE: Work Package Leader (Knowledge Transfer)
OBJECTIVES: Determining the epidemiological and contextual nature of spontaneous preterm birth in three sub-Saharan African countries (Kenya, Nairobi and Mozambique), while developing technical infrastructure and training research scientists.
SCIENTIFIC CONTRIBUTION: Alessia Visconti designed and delivered workshops, rated as “exceptional” by participants, on basic and advanced skills (*e.g.*, version control and collaboration using Git/GitHub, Bash programming, and metagenomic data analysis).
- 2020 - 2021 TITLE: A multi-omics study to dissect the role of the gut microbiome in IgA nephropathy risk
FUNDER: King’s College London - Peking University Health Science Centre Joint Institute for Medical Research; BUDGET: £74,000
COLLABORATORS: Peking University, China
ROLE: Work package leader (data analysis).
OBJECTIVES: Identification of genes and microbes associated with IgAN or with IgA glycosylation.
SCIENTIFIC CONTRIBUTION: Alessia Visconti conducted several association studies, including one of the first GWAS of IgA glycosylation [J56].
- 2019 TITLE: Studying the microbiome-disease connection
FUNDER: King’s Undergraduate Research Fellowship (KURF); BUDGET: £2,000 (£1,000 for a student bursary and £1,000 for laboratory costs)
ROLE: PI
OBJECTIVES: Exploratory analysis aimed at assessing the effect of chronic diseases on the taxonomic and functional composition of the gut microbiome.
SCIENTIFIC CONTRIBUTION: Alessia Visconti supervised a summer internship (4 weeks, 150 hours), which aimed at generating preliminary results to explore a potential new research direction.
- 2018 TITLE: Studying the microbiome-immune system interplay
FUNDER: King’s Undergraduate Research Fellowship (KURF); BUDGET: £3,000 (£2,000 for a student bursary and £1,000 for laboratory costs)
ROLE: PI
OBJECTIVES: Exploratory analysis aimed at assessing the interactions between the taxonomic and functional composition of the gut microbiome and immune-system composition (immunophenotypes).
SCIENTIFIC CONTRIBUTION: Alessia Visconti supervised a summer internship (8 weeks, 300 hours), which aimed at generating preliminary results to explore a potential new research direction.
- 2017 TITLE: Metagenome-Wide Association Study of Lipid Profiles in Twins
FUNDER: King’s Undergraduate Research Fellowship (KURF); BUDGET: £3,000 (£2,000 for a student bursary and £1,000 for laboratory costs)
ROLE: PI
OBJECTIVES: Exploratory analysis aimed at assessing associations between the taxonomic and functional composition of the gut microbiome and systemic lipid profiles (LDL and HDL).
SCIENTIFIC CONTRIBUTION: Alessia Visconti supervised a summer internship (8 weeks, 300 hours), which aimed at generating preliminary results to explore a potential new research direction.

2016 - 2018 TITLE: A high-resolution map of copy number and structural variation in Qatari genomes and their contribution to quantitative traits and diseases
 FUNDER: Qatar Foundation; BUDGET: £160,521
 COLLABORATORS: Sidra Medical and Research Centre, Qatar
 ROLE: Work package leader (data analysis).
 OBJECTIVES: Development and validation of an *ensemble* approach for the identification of structural variants (deletions, duplications, inversions, translocations, and insertions) and characterisation of the mechanisms through which these variants influence disease susceptibility.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti was the scientific lead for the statistical and computational analyses in several publications [J34, J54, J63] and co-supervised a PhD student (Niccolò Rossi), continuing to do so throughout his postdoctoral period.

PROJECTS TO WHICH ALESSIA VISCONTI PARTICIPATES AS RESEARCHER

Alessia Visconti has a prominent role in all projects, as shown by the number of publications in which she is first/last author.

2023 - PRESENT TITLE: Repertor.IO: Incorporating Patient Preference Studies into Clinical Research and Decision Model
 FUNDER: *Nord Ovest Digitale e Sostenibile (L'ecosistema dell'innovazione)*; funded by the Italian Ministry of University and Research (MUR) under the National Recovery and Resilience Plan (PNRR)
 COLLABORATORS: University of Padua, Italy
 OBJECTIVES: Developing a platform for the collection, annotation, and meta-analysis of patients preference studies.
 SCIENTIFIC CONTRIBUTION: After exploring, through a survey, the use of patient preferences in clinical care and decision making [J60], Alessia Visconti contributed to the development of the meta-analysis framework, to the activities leading to the creation of the start-up (see the TECHNOLOGY TRANSFER AND PUBLIC ENGAGEMENT section for details), and to the preparation of funding applications (€28,000 from Finpiemonte). As part of this project, she co-supervises a PhD student (Matteo Bracco).

2023 - 2026 TITLE: Impact of plant diversity on cardiometabolic health
 FUNDER: Chronic Disease Research Foundation (CDRF)
 COLLABORATORS: King's College London, UK
 OBJECTIVES: Assess how habitual diet influences the gut microbiota and the metabolites it produces and, in turn, cardiometabolic disease risk.
 RESEARCH CONSUMPTION: Alessia Visconti was responsible for the preparation of meta-genomic data, and contributed to the statistical analysis of multi-*omics* and dietary data in the study described in [J61].

2023 - 2025 TITLE: TrustAlert
 FUNDER: Fondazione Compagnia San Paolo e Fondazione Cassa Depositi e Prestiti
 COLLABORATORS: Bruno Kessler Foundation, Italy, Links Foundation, Italy, Innovo Group (now DedaNext), Italy, Local Health Authority Alba-Bra (ASL-CN2), Italy, Cottolengo Hospital, Italy
 OBJECTIVES: Developing AI-based solutions for (a) near-real-time identification of health emergencies, (b) mapping vulnerable and multimorbid populations, and (c) crate a *living lab* for micro and macro simulations.
 RESEARCH CONTRIBUTES: Alessia Visconti was involved in (a) supporting the development of an event-detection system based on the GDELT platform for identifying and tracking the evolution of public health emergencies [J62], (b) analysing administrative healthcare data to predict short-term readmission (< 90 days) using deep learning approaches [P6], and (c) analysing administrative healthcare data to identify population multi-morbidity patterns, using both a Convolutional Autoencoder (ConvAE) architecture and a Top2Vec-based approach. As part of this project, she co-supervised two PhD students (Emanuele Koumantakis, Sara De Luca), four research assistants (Matteo Bracco, Giuseppe Martino, Emanuele Pietropaolo, Michela Bersia), and two MSc students (Francesca Rondinone, Lorenzo Miglietta).

- 2021 - 2022 TITLE: Predicting Response to Immunotherapy for Melanoma with gut Microbiome and metabolomics - The PRIMM Study
 FUNDER: Seerave Foundation
 COLLABORATORS: University Medical Center, Groningen, The Netherlands and University of Trento, Italy
 OBJECTIVES: Use multi-*omics* data to predict which patients with melanoma would benefit from the use of immune checkpoint inhibitors.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti conducted the analysis for the identification of glyco-markers [J45], and contributed to the analysis of proteomic data [J42]. As part of this project, Alessia Visconti co-supervised a PhD student (Karla Lee) and a postdoctoral researcher (Niccolò Rossi) on the statistical analysis of multi-*omics* data.
- 2020 - 2022 TITLE: ZOE COVID Study
 FUNDER: Department of Health and Social Care (DHSC)
 COLLABORATORS: ZOE Ltd (King's College London start-up), Massachusetts General Hospital, US
 OBJECTIVES: Monitor the SARS-CoV-2 pandemic in near real time to optimise resource allocation and assess the effectiveness of prevention and treatment measures.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti collaborated within the ZOE COVID Study (now ZOE Health Study) to develop the pipeline for analysing data submitted daily by approximately 4 million users [J39]. She also led two studies on the cutaneous symptoms of the infection [J32, J41], and collaborated as data analyst in several other studies [J25, J27, J29, J31, J33, J36].
- 2019 - 2021 TITLE: Dissecting the mechanisms of immune-mediated inflammation: a systems-immunology approach
 FUNDER: Medical Research Council (MRC)
 OBJECTIVES: Identifying targetable networks of circulating immune cells and cytokines involved in the normal immune-mediated inflammation process, and identify those disrupted in immune-mediated inflammatory diseases.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti was the leading researcher for the bioinformatics analyses aiming at (a) the reverse engineering of immune cell co-expression networks and their involvement in a set of autoimmune diseases, and (b) the identification of genetic variations and microbes/metabolites responsible for the development of such diseases. As part of this project, she supervised a postdoctoral researcher (Niccolò Rossi).
- 2019 - 2020 TITLE: Identification of multi-disease drug targets through systems-immunology dissection of immune ageing
 FUNDER: Sanofi iAwards Europe 2019
 OBJECTIVES: Identify a set of immune biomarkers for the early identification of individuals at higher risk of accelerated immune ageing.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti was the scientific lead for the statistical analysis, and supervised a postdoctoral researcher (Niccolò Rossi) on machine-learning topics.
- 2018 - 2020 TITLE: Industry-sponsored research collaboration
 FUNDER: Danone Nutricia Research
 OBJECTIVES: Study the role of the gut microbiome in modulating the effects of yoghurt consumption on human health.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti collaborated to the statistical analysis, and was responsible for extracting sequence information from metagenomic data [J40].
- 2016 - 2018 TITLE: Gut microbiome modulation of fasting glucose homeostasis and postprandial glycaemic response in TwinsUK and PREDICT: towards personalised diet for healthy aging
 FUNDER: Chronic Disease Research Foundation (CDRF)
 COLLABORATORS: University of Trento, Italy
 OBJECTIVES: Studying the influence of the gut microbiome on cardiometabolic health.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti (a) developed a tool for the reproducible analysis of metagenomic data (YAMP) [J16], (b) performed the bioinformatics analysis of metagenomics and metabolomics data [J21, J30], (c) collaborated to several studies exploring the influence of the gut metagenome on human health [J47, J48, J49].

- 2014 - 2016 TITLE: An integrative genomics approach for non-invasive diagnostic biomarkers discovery in IgA nephropathy
 FUNDER: Medical Research Council (MRC)
 COLLABORATORS: Imperial College London, UK
 OBJECTIVES: Identify glycomics and genetic biomarkers for improving IgAN diagnosis and treatment.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti applied statistical and bioinformatics approaches for studying the role of IgA and its glycosylation profiles in the development of IgA nephropathy [J9, J37, J56].
- 2013 - 2018 TITLE: Genomic analysis of Type 2 Diabetes in Qatar, towards diabetes personalized medicine
 FUNDER: Qatar Foundation
 COLLABORATORS: Weill Cornell Medicine-Qatar, Qatar
 OBJECTIVES: Identifying the genetics and epigenetics basis of Type 2 Diabetes.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti developed and implemented an approach for the association testing of quantitative traits in related individuals (*PopPanTe*) [J12], and conducted the bioinformatics analysis for [J7] and [J13].
- 2013 - 2015 TITLE: Senescence and melanoma – An integrative systems biology approach to characterise the link between reduced biological senescence and melanoma susceptibility
 FUNDER: British Skin Foundation
 COLLABORATORS: Hospital Clinic of Barcelona, Spain, QIMR Berghofer Medical Research Institute, Australia, Beijing Institute of Genomics, China, Institute of Cancer and Pathology, UK, Erasmus MC University Medical Center, The Netherlands, Women’s Hospital, US
 OBJECTIVES: Identifying the genetics basis of melanoma skin cancer and of its risk phenotypes.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti applied statistical and bioinformatics approaches for studying melanoma, melanoma risk phenotypes, and their connection with ageing [J10, J11, J14, J15, J18, J20, J23, J24, J28]. As part of this project, she co-supervised a PhD student (Marianna Sanna).
- 2012 - 2013 TITLE: LIMPET – Isotropic And Anisotropic Lipophilicity To Model Permeability Of New Therapeutic Peptides
 FUNDER: Compagnia di San Paolo
 OBJECTIVES: Applying machine Learning approaches to predicts peptides permeability on the basis of experimental non-conventional lipophilicity indexes.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti (*a*) evaluated the ability of some combinations of descriptors/algorithms to find the best model to predict the lipophilicity of small peptides [J5], and (*b*) performed the bioinformatics analyses to assess the ability of more than 200 compounds to act as hydrogen-bond donors [J3].
- 2007 - 2011 TITLE: BioBITs – Developing white and green biotechnologies by converging platforms from biology and information technology towards metagenomics
 FUNDER: Regione Piemonte, *Bando Converging technologies 2007*
 COLLABORATORS: University of Eastern Piedmont, Italy, Consiglio Nazionale delle Ricerche - Istituto Protezione Piante, Centro Colture Sperimentali Valle d’Aosta Srl, Isagro Ricerca Srl, Etica Srl
 OBJECTIVES: Identifying and characterising populations of uncultivable bacteria living inside a symbiotic, arbuscular mycorrhizal fungus.
 SCIENTIFIC CONTRIBUTION: Alessia Visconti contributed to the development of a modular framework for the analysis of metagenomics sequences, and was responsible for the co-clustering module [BC1].

RESEARCH ACTIVITIES NOT SUPPORTED BY DIRECT FUNDING

- 2025 - PRESENT TITLE: Use of generative AI for teaching
 ROLE: co-PI
 OBJECTIVES: Development of a virtual patient simulation environment through which medical students can practise in realistic clinical scenarios in a safe environment.

SCIENTIFIC CONTRIBUTION: Alessia Visconti co-coordinates the design and development of an interactive chatbot based on a multi-agent architecture. As part of this project, she co-supervised two MSc students (Boris Turcan, Samuele Perrotta) and a medical student (Mohamed Ayadi).

2024 - PRESENT **TITLE:** Multi-*omics* data integration
ROLE: co-PI
OBJECTIVES: Development of new methodologies for patient stratification based on their multi-*omics* profiles
SCIENTIFIC CONTRIBUTION: Alessia Visconti co-coordinated the development of a novel three-step tool which successfully stratified both synthetic patients and those with chronic myeloid leukaemia. As part of this project, she co-supervises a PhD student (Sara De Luca).

PARTICIPATION IN RESEARCH GROUPS

2024 - PRESENT **Co-coordinator** of the Working Group on *Machine learning in clinical research* of the Italian Society of Medical Statistics and Clinical Epidemiology (SISMEC).
2016 - 2022 **Researcher** within the TREatment of ATopic eczema (TREAT) Registry Taskforce (a consortium involving more than 30 research institutes in 9 countries)
PUBLICATIONS: [J38, J50, J58].

AWARDS

JULY 2018 **Winner of one of the challenges at the *BioDataHack 2018 - Genomic, Biodata and Improving Health Outcomes***, a competition organised by the Wellcome Trust (2-3 July 2018, Wellcome Genome Campus, Hinxton, UK). The challenge, undertaken in collaboration with Jun Aruga, Oliver Giles, Ioannis Valasakis, and Chen Zhang, required the development of an innovative solution to answer the question “*How can we use mobile technology to transform biological data processing?*”. The proposed solution was then assessed on scientific and methodological quality as well as creativity, communication skills, and ability to work as part of a team.
Alessia Visconti’s team proposed the development of a device for continuous monitoring of the gut microbiome in patients with inflammatory bowel disease (IBD). The device integrated a metagenomic analysis pipeline developed by Alessia Visconti (*YAMP*) [J16] with a shallow neural network to classify healthy individuals and patients with one of the two main forms of the disease, Crohn’s disease and ulcerative colitis. The device then used a multivariate linear regression model to derive a quantitative score of disease progression, enabling the near-real-time monitoring of the effects of lifestyle changes or pharmacological treatment. During the two-day hackathon, the team successfully ported *YAMP* to Arm’s 64-bit architecture, where the data could be processed within a few hours, showing that the analysis of microbial data can be successfully performed outside centralised data centres. Finally, the team showed that the proposed approach was able to separate patients with IBD from healthy controls with more than 90% accuracy. The next step would be developing models offering personalised advice and treatments based on interventions that have been effective in patients with similar microbiome profiles.

JAN 2012 **Awarded two years postdoctoral fellowship** by the Italian Minister of Education, University and Research (MIUR; duration: January 2012 - December 2013).

JAN 2012 **Awarded a postdoctoral training grant** (€3,000) by the Regione Piemonte.

JUN - OCT 2010 **Winner of one of the challenges presented in the *DREAM5 – Network Inference Challenge***, an international competition organised by the Dialogue for Reverse Engineering Assessments and Methods (DREAM) project to benchmark tools for the inference of gene regulatory networks from gene expression data. Methods were evaluated on simulated datasets and real data from *E. coli* and *S. cerevisiae*. Performance was assessed as a binary classification problem (regulatory edge present/absent) using reference (*gold-standard*) networks and standard metrics (AUROC and AUPR).

The approach developed by Alessia Visconti together with Roberto Esposito and Francesca Cordero [P4, J2], which integrated established statistical methods for gene regulatory network inference into a Naive Bayes classifier, was compared with 35 other methods, ranking 3rd overall on real biological data and 1st in the reconstruction of the *S. cerevisiae* gene regulatory network.

JAN 2009 **Awarded three years PhD fellowship** by the Italian Minister of Education, University and Research (MIUR; duration: January 2009 - December 2011).

REVIEW ACTIVITY

EDITORIAL BOARD MEMBERSHIP Since September 2019, Alessia Visconti is *Review Editor* of Human and Medical Genomics (speciality section of Frontiers in Genetics).

REVIEWER Alessia Visconti serves as a reviewer for several international journals, including Acta Dermato-venereologica, Bioinformatics, BioData Mining, BMC Cancer, BMC Nutrition, BMC Supplements, British Journal of Cancer, Computational and Structural Biotechnology Journal, Communications Biology, Digital Health, European Journal of Human Genetics, Frontiers in Cellular and Infection Microbiology, GigaScience, Gut Microbes, Journal of the European Academy of Dermatology and Venereology, Knowledge and Information Systems (KAIS), Machine Learning, PLoS Computational Biology, PLoS One, Nature Communications, Scientific Reports, Journal of the Royal Society Interface.

INVITED TALKS

APR 21ST, 2026 “*It takes a village to raise a scientist*” at the COMputer SciEnce DeparTmEnt (COMETE) 2026 PhD Workshop, Turin, Italy

NOV 7TH, 2025 “*AI nella sanità pubblica: un approccio interpretabile per la gestione dei pazienti*” at the Intelligenza Artificiale, Macchine e Futurismo, opening day of the MSc in *Data analysis for the business intelligence and data science*, Turin, Italy

JUN 17TH, 2019 “*Reproducible shotgun metagenomic analysis with Nextflow and containers*” at the 1st London Bioinformatics Frontiers Frontiers conference, London, UK

NOV 23RD, 2018 “*Nextflow on the go*” at the 2nd Nextflow workshop, Barcelona, Spain

DEC 6TH, 2017 “*YAMP: a framework enabling reproducibility in metagenomics research*” at the Computational Biology seminar series, Francis Crick Institute, London, UK

DEC 13TH, 2017 “*My reproducible adventure in (meta)genomics (A tale of workflow development, research reproducibility & open science)*” at the Research/hers code seminar series, London, UK

SEP 15TH, 2017 “*Simplifying shotgun metagenomics analysis with Nextflow*” at the 1st Nextflow workshop, Barcelona, Spain

JUL 7TH, 2012 “*Knowledge-driven Co-clustering of Gene Expression Data*” at the Center for Biological Sequence Analysis, Technical University of Denmark, Lyngby, Denmark

MAY, 8TH 2009 “*Using a priori knowledge for the reverse engineering of gene regulatory networks*” at the Computer Laboratory, Cambridge University, Cambridge, UK

TECHNOLOGY TRANSFER AND PUBLIC ENGAGEMENT

Alessia Visconti's activities include, in addition to the development and release of open-source software (see the DEVELOPMENT OF RESEARCH SOFTWARE section), the initiatives listed below.

- JUN 2026 - PRESENT **Selected for Funds TOgether**, the crowdfunding programme of the University of Turin dedicated to high-impact and innovative projects with social and environmental impact. Alessia Visconti (PI of the project) and her team (Paola Berchiarella, Barbara Sini, Matteo Bracco) passed a selection process assessing both the quality of the project proposal, including its scientific value and innovativeness, and the suitability of the project objectives for crowdfunding. The proposed project aims at understanding how climate change affects psychological wellbeing by developing a citizen-science app to collect longitudinal data on psychological wellbeing and integrate them with climate and environmental data. The aim is to create the first participatory observatory on the relationship between climate and mental health, raising public awareness and supporting concrete interventions at the local level.
- SEP 2025 - PRESENT **Participation in the *Penne Amiche della Scienza* project.** The project follows the format of "*Letters to a Pre-Scientist*" and aims at introducing primary and lower-secondary school students to the world of science through an exchange of letters with researchers. These letters provide an opportunity to share researchers' educational and research journeys, answer students' questions, and stimulate their interest in scientific disciplines. Alessia Visconti participated in the 2025-26 edition of the programme, which involved sending four letters (one introductory letter and three responses) and visiting the participating school (*Scuola Secondaria I grado Antonio Gramsci, Class 2A*). She has also enrolled in the 2026-27 edition and will be paired with a school in September 2026.
- DEC 2025 - MAR 2026 **Recording of a video lecture** (in Italian) entitled "*Introduction to Large Language Models*" (Module 3) within the transdisciplinary Massive Open Online Course (MOOC): "*Artificial Intelligence in Veterinary Medicine: Perspectives and Applications*", which was funded by the National Recovery and Resilience Plan (PNRR), Mission 4, Education and Research.
- JUL 2025 - PRESENT **Establishment of the start-up "Repertor.IO"**, approved by the Department Council on June 11th, 2024 and established on July 14th, 2025. Repertor.IO was recognised as an academic spin-off by the Board of Directors of the University of Turin on July 18th 2025. Alessia Visconti's role is *Chief Research Communication Manager*.
- JUL 2017 – FEB 2023 **Member and mentor** of the *Artificial Intelligence Club for Gender Minorities*, which aimed at promoting gender diversity in the artificial intelligence and scientific community via meetups and mentorship. Alessia Visconti organised workshops on collaborative data science using Git and GitHub. From May 2018, she also co-organised the group monthly journal club.
- MAR 2016 – SEP 2017 **Member and mentor** of the *RLadies London* community and of the *Research[her]* Research community. These groups aim at promoting gender diversity in the R and STEM community via meetups, mentorship and global collaboration. With *Research[her]*, Alessia Visconti gave speeches on reproducibility and workflow development.

DEVELOPMENT OF RESEARCH SOFTWARE

Alessia Visconti is the sole developer unless otherwise specified. The software is distributed under an open-source licence and is available at <https://github.com/alessia> or upon request.

AID-ISA biclustering algorithm leveraging *a priori* knowledge [J4]

CDoT	exact algorithm for answering <i>Maximum a Posteriori</i> queries on tree-structured probabilistic graphical models [P5] (developed in collaboration with Roberto Esposito)
FAMCNV (v2.0)	algorithm for testing the association between copy-number variation (CNV) and quantitative phenotypes in families (developed in collaboration with Mario Falchi)
GOCLUST	co-clustering algorithm that leverages biological information extracted from the Gene Ontology to improve the quality of the results [BC2, P2]
MOTIFSLINKER	algorithm for extracting protein motifs and grouping them into families using a constrained co-clustering approach [P1]
PICNIC	tool for searching topologies within chemical molecules, which has been used to support teaching at the Department of Molecular Biotechnology and Health Sciences, University of Turin
POPPANTE	algorithm for pairwise association testing of quantitative data in related samples [J12]
RGO	algorithm for grouping Gene Ontology terms according to their function to obtain a more compact and informative ontology [J1, P3]
SPOT	algorithm for the exhaustive search of frequent motifs in biological sequences [T2]
YAMP	reproducible pipeline for the analysis of metagenomic data [J16]

SERVICE ACTIVITIES

MAR 2026 - PRESENT	Mentor in the <i>Peer Mentoring for Innovation and Excellence in Learning and Teaching</i> programme. The initiative, promoted by the Teaching and Learning Center of the University of Turin, aims at supporting continuous improvement in university teaching through structured peer-to-peer exchange. Faculty members serving as mentors support colleagues (the mentees) in reflecting on their teaching practices through individual meetings, reciprocal classroom observations, and feedback sessions, promoting the sharing of experiences, the adoption of innovative teaching methods, and the development of an academic community committed to teaching quality. In March 2026, Alessia Visconti, together with two other mentors, supported one mentee from the Department of Drug Science and Technology, University of Turin.
DEC 2025 - PRESENT	Appointed member of the Steering Committee of the Interdepartmental Research Centre for Service Learning representing the Department of Clinical and Biological Sciences, University of Turin. The Interdepartmental Centre aims to promote research, training, and educational innovation for the development and dissemination of <i>service learning</i> , a pedagogical approach that integrates educational objectives with community service activities.
NOV 2025 - PRESENT	Member of the Board of the Medical Residency program in <i>Medical Statistics and Biometrics</i> , University of Turin.
NOV 2024 - PRESENT	Member Scientific Board for the PhD program in <i>Complex Systems for Quantitative Biomedicine</i> , University of Turin.
DEC 2023 - PRESENT	Department Internal Staff Contact (DISC) for the Department of Clinical and Biological Sciences. She is responsible for managing the RedCAP platform of the School of Medicine at the University of Turin (MEDCAP).
DEC 2019 – SEP 2023	Co-organiser of a series of workshops aiming at teaching basic computer literacy and data analysis skills with the King’s College London (KCL) Carpentries.
NOV 2016 – AUG 2022	Co-organiser of the Journal Club on “ <i>Genetic and epigenetic regulation of gene expression</i> ” at the Department of Twin Research & Genetic Epidemiology, King’s College London.

JAN 2012 – DEC 2013	Elected member of the Department Council at the Department of Computer Science, University of Turin, representing fixed-term researchers.
JAN 2009 – DEC 2011	Elected member of the Department Council at the Department of Computer Science, University of Turin, representing PhD students.

TEACHING ACTIVITY

TO BSc AND MSc STUDENTS

A.Y. 2025-26 - PRESENT	Co-Lecturer for the “<i>Multimorbidity</i>” elective teaching activities for the <i>Faculty of Medicine and Surgery - MedInTO</i>, Department of Clinical and Biological Science, University of Turin. The session led by Alessia Visconti (2h) explains how computational medicine is used to predict the clinical trajectories and to stratify patients with multimorbidity. Language: English. Lecturer for the “<i>Introduction to Workflow Orchestration Managers</i>” workshop for the <i>Advanced Bioinformatics</i> module, MSc in Genomic Medicine, King’s College London. The workshop (3h) follows the “live coding” principle Alessia Visconti champions and offers a primer on Nextflow, a very popular workflow orchestration manager. Language: English.
A.Y. 2024-25 - PRESENT	Lecturer for the “<i>Medical statistics</i>” module within the “<i>Evidence-based nursing</i>” course, Nursing School, Department of Clinical and Biological Science, University of Turin. The module (15h) covers the fundamentals of biostatistics and is run in four parallel sessions in two different university campuses, Cuneo (from a.y. 2024-25) and Orbassano (from a.y. 2026-27). Language: Italian. Lecturer for the “<i>Medical statistics</i>” module within the “<i>Preparatory to research</i>” course, Degree in Psychiatric Rehabilitation Techniques, Department of Clinical and Biological Science, University of Turin. The module (24h) covers the fundamentals of biostatistics through lectures and practical activities. Language: Italian. Co-lecturer for the “<i>Leveraging Large Language Models for hands-on learning in medical education and research</i>” elective teaching activities for the <i>Faculty of Medicine and Surgery - MedInTO</i>, Department of Clinical and Biological Science, University of Turin. The activity (10h) is organised as a set of lectures, hands-on activities, and discussions. It covers how large language models work, their hidden costs, and whether to use them in reading, writing, studying, and in doing clinical research and work. Language: English.
A.Y. 2024-25	Co-lecturer for the “ <i>Methodology of social-health research</i> ” module within the “ <i>Sociology and Methodology of Health and Social Care Research</i> ” course, Degree in Professional Education, University of Turin. The module (20h) covered the fundamentals of biostatistics. Language: Italian.
A.Y. 2013-14	Teaching assistant for the “<i>Human Molecular Genetics</i>” MSc Department of Genomics of Common Diseases, Imperial College London. Alessia Visconti supported students during the practical sessions within the following courses: - The Unix Shell - R Programming - Exploratory Data Analysis and Probability - Quantitative genetics - Next Generation Sequencing Data Analysis
A.Y. 2013-14	Teaching assistant for the “ <i>Data analysis</i> ” course, Department of Biological Science, University of Turin. Alessia Visconti supported students during the practical sessions on R programming (10h) as well as during one-to-one meetings. Language: Italian.

- A.Y. 2012-13 & 2013-14 **Teaching assistant** for the “*Operating System*” course, Department of Computer Science, University of Turin. Alessia Visconti supported students during the practical sessions (25h) as well as during one-to-one meetings. Language: Italian.
- A.Y. 2012-13 **Lecturer** for the “*Operating System and Networking*” course, Interfaculty School of Strategic Studies, University of Turin. Alessia Visconti was responsible for the practical sessions covering the basis of GNU/Linux, the Unix shell, and process management (28h). She designed the final project, *i.e.*, the development of a basic client/server application in C. Language: Italian.
- A.Y. 2010-11 & 2011-12 **Teaching assistant** for the “*Database*” course, Department of Computer Science, University of Turin. Alessia Visconti supported students during the practical sessions (25h) as well as during one-to-one meetings. Language: Italian.
- A.Y. 2010-11 **Teaching assistant** for the “*Formal Language*” course, Department of Computer Science, University of Turin. Alessia Visconti supported students during the practical sessions (25h) as well as during one-to-one meetings. Language: Italian.
- Teaching assistant** for the “*Statistics and data mining with SAS*” course, Department of Mathematics, University of Turin. Alessia Visconti prepared all the course online material, including a self-evaluation questionnaire, and recorded 3 hours of video lessons on SAS Enterprise Miner. Language: Italian.
- A.Y. 2009-10 **Lecturer** for the “*Computer Science*” course, Department of Letters and Philosophy, University of Turin. The hands-on course (30h in two parallel sessions) covered the MS Office suite. Language: Italian.
- A.Y. 2005-06 & 2006-07 **Teaching assistant** for the “*Program Languages - JAVA*” course, Department of Computer Science, University of Turin. Alessia Visconti supported students during the practical sessions. Language: Italian.
- A.Y. 2004-05 **Teaching assistant** for the “*Program Languages - C*” course, Department of Computer Science, University of Turin. Alessia Visconti supported students during the practical sessions. Language: Italian.

TO MEDICAL RESIDENTS

- A.Y. 2025-26 – PRESENT **Co-lecturer** for the “*Introduction to biostatistics*” course for medical residents at University of Turin. The course (7h) covers the fundamentals of biostatistics. Language: Italian.
- A.Y. 2024-25 – PRESENT **Co-lecturer** for the “*Integrating Large Language Models into clinical work and research: a critical approach*” course for medical residents at University of Turin. The course (7h) is organised as a set of lectures, hands-on activities, and discussions. It covers how large language models work, their hidden costs, and whether to use them in the clinic and in clinical research. Language: English.

TO POSTGRADUATE AND PHD STUDENTS, AND POSTDOCTORAL RESEARCHERS

- A.Y. 2025-26 – PRESENT **Lecturer** for the “*Introduction to biostatistics*” workshop for the PhD program in Complex Systems for Quantitative Biomedicine, at the University of Turin. The workshop (2h) covers, through practical exercises, the fundamentals of biostatistics. Language: Italian.

Co-lecturer for the “*Machine Learning in Clinical Research: Methods, Interpretability, and Limitations*” workshop for the Advanced Master programme in Epidemiology, Department of Medical Sciences, University of Turin. The workshop (8h) offers a primer on supervised and unsupervised machine learning, and is organised as a set of lectures with a final hands-on activity. Language: Italian.

A.Y. 2024-25 – PRESENT **Lecturer** for the “*Introduction to Git*” workshop for the PhD program in Complex Systems for Quantitative Biomedicine, at the University of Turin. The workshop (8h) is centred around the idea of “live coding”, where attendees code along with the instructors thus getting useful hands-on experience and improving their ability to explore the topics on their own. Language: English.
Lecturer for the “*A Crash Course on Machine Learning for Beginners: Practice*” workshop for the PhD program in Complex Systems for Quantitative Biomedicine, at the University of Turin. The workshop (8h) follows the same “live coding” principle, and offers an engaging hands-on session on supervised learning. Language: English.

A.Y. 2019-20 TO 22-23 **Instructor** for several programming and data analysis workshops held at King’s College London covering the following topics:

- The Unix Shell
- Version Control and collaboration with Git/GitHub
- Python Programming
- R Programming
- Introduction to Working with Data
- OpenRefine

These workshops were designed for PhD students and early career researchers but were open to researchers at every level, including PIs, and offered regularly (roughly twice a year). They have always been evaluated as “excellent” or “exceptional” by the attendees. The workshops are centred around the idea of “live coding”. Language: English.

A.Y. 2021-22 **Instructor** for “*The Unix Shell*” and the “*Version Control and collaboration with Git/GitHub*” workshops (18h) for students and researchers of the MRC Gambia and the AKU Nairobi Research Centres. Both workshops were rated as “exceptional” by the attendees. Language: English.

Instructor for the “*Metagenomics Data Analysis: Investigating the invisible world of microbes*”, Department of Twins Research & Genetic Epidemiology, King’s College London. The entry-level workshop (14h), which included frontal lessons and hands-on sessions, was designed for researchers at every level (including but not limited to PhD students, post-doctoral researchers, and PIs). Language: English.

OUTSIDE ACADEMIA

A.Y. 2021-22 **Instructor** for a “*Version Control and collaboration with Git/GitHub*” workshop at the UK Health Security Agency (UKHSA, 9h).

SUPERVISION ACTIVITY

BSC & MSc THESES

A.Y. 2025-26 **Supervisor** of Mr Lorenzo Miglietta’s MSc thesis in *Computer Engineering - Artificial Intelligence and Data Science* at the Polytechnic of Turin, Italy. Thesis title: *Explainable patients’ trajectory prediction with Healthcare Administrative Data*.

Supervisor of Ms Triparna Day’s MSc thesis in *Biotechnological and Chemical Sciences in Diagnostics* at the University of Turin, Italy. Thesis title: *Assessing the Accuracy of Computational Taxonomic Inference the Low-Diversity Vaginal Microbial Ecosystems*.

Co-supervisor of Mr Mohamed Ayadi's thesis in *Medicine and Surgery* at the University of Turin, Italy. Thesis title: *Valutazione dell'efficacia di un sistema di simulazione clinica immersiva basato su Intelligenza Artificiale nel miglioramento delle competenze cliniche e non tecniche negli studenti di Medicina.*

Co-supervisor of Mr Samuele Perrotta's MSc thesis in *Computer Science* at the University of Turin, Italy. Thesis title: *MEDtrAIner: un sistema Multi-Agente per la simulazione di scenari clinici a supporto della didattica medica.*

Co-supervisor of Mr Boris Turcan's MSc thesis in *Computer Science* at the University of Turin, Italy. Thesis title: *MEDtrAIner Plus Una nuova estensione per la simulazione di scenari clinici complessi e realistici.*

A.Y. 2024-25 **Co-supervisor** of Ms Francesca Rondinone's MSc thesis in *Stochastics and Data Science* at the University of Turin, Italy. Thesis title: *Explainable convolutional autoencoder for unsupervised patient phenomapping with Healthcare Administrative Data.*

A.Y. 2021-22 **Co-supervisor** of Ms Darvina Magandran's MSc thesis in *Microbiome in Health and Disease* at King's College London. Thesis title: *The role of the microbiome on atopic dermatitis.*

A.Y. 2020-21 **Co-supervisor** of Ms Raphaela Joos's MSc thesis in *Microbiome in Health and Disease* at King's College London. Thesis title: *Impact of probiotic Lactobacillaceae on Lipid Metabolism.*

Co-supervisor of Ms Petra Blackburn's MSc thesis in *Microbiome in Health and Disease* at King's College London. Thesis title: *Investigating the association between the microbiome and gene expression in adipose tissue.*

Co-supervisor of Ms Natalie Falshaw's MSc thesis in *Microbiome in Health and Disease* at King's College London. Thesis title: *Investigating the association between the microbiome and gene expression in adipose tissue.*

A.Y. 2019-20 **Co-supervisor** of Ms Xinyu Huang's MSc thesis in *Pharmacology* at King's College London. Thesis title: *Impact of proton pump inhibitors on composition and metabolism of gut microbiome.*

A.Y. 2018-19 **Co-supervisor** of Ms Miriam Margari's MSc thesis in *Genomic Medicine* at Imperial College London. Thesis title: *Inference of parent-of-origin effects using human gene expression data.*

A.Y. 2017-18 **Co-supervisor** of Dr Robin Mesnage's MSc thesis in *Bioinformatics* at Birkbeck University of London. Thesis title: *A Metagenome-wide association study suggests that glycome composition associates with pathogenic bacteria abundance in the gut microbiome.*

A.Y. 2010-11 **Co-supervisor** of Mr Marco Gallizio's BSc thesis in *Computer Science* at the University of Turin, Italy. Thesis title: *A web interface for querying the Restructured Gene Ontology.*

POST-MASTER STUDENTS

A.Y. 2019-20 **Co-supervisor** of Mr Simon Couvreur in his PhD rotation project at King's College London.

Co-supervisor of Ms Helen King in her PhD rotation project at King's College London.

PHD STUDENTS

2025 - PRESENT **Supervisor** of Mr Matteo Bracco in his PhD project at University of Turin. Publications: [J60].

2024 - PRESENT **Co-supervision** of Ms Sara De Luca in her PhD project at Campus Bio Medico University of Rome. Publications: [J62].

Co-supervision of Mr Emanuele Koumantakis in his PhD project at University of Turin. Publications: [J59, PP2].

- 2021 - 2023 **Co-supervisor** of Mr Roger Compte Boixader in his PhD project at King's College London. Thesis title: *Genetic and multi-omics determinants of intervertebral disk degeneration and their intersection with chronic pain*. Publications: [J52].
- 2019 - 2024 **Co-supervisor** (unofficial) of Ms Xinyuan Zhang in her PhD project at King's College London. Thesis title: *Interplay between drug and gut microbiome*. Publications: [J43, J57, J61, J64].
- 2020 - 2022 **Assistance with the supervision** of Ms Karla Lee in her PhD project at King's College London (supervision limited at the multi-omics data analyses. Thesis title: *Dissecting the Role of the Gut Microbiome in Advanced Melanoma*. Publications: [J42, J45].
- 2016 - 2019 **Co-supervisor** (unofficial) of Mr Niccolò Rossi during his research visit at King's College London (Visiting PhD Student from the University of Lisboa). Thesis title: *Tackling the molecular basis of lipid metabolism: from candidate genes testing in a disease cohort to multi-omics approaches in unselected populations*. Publications [J34, J54].
- 2018 - 2020 **Supervisor** of Ms Giulia Piaggese during her research visit at King's College London (Visiting PhD Student from the University of Turin). Thesis title: *Evaluation of DNA methylation in leukocyte subpopulations associated with tobacco smoking exposure*. Publications: [J35].
- 2017 - 2018 **Assistance with the supervision** of Ms Chiara Sacco during her research visit at King's College London (Visiting PhD Student from the University of Bologna). Thesis title: *Statistical Methods for the Detection of the Parent-of-Origin Effect in Genome-Wide Genotype Data*.
- 2014 - 2020 **Co-supervisor** (unofficial) of Ms Marianna Sanna in her PhD project at King's College London and her research activity at Imperial College London. Thesis title: *Omics studies in Skin for ageing and cutaneous melanoma risk factors*. Publications: [J10, J15, J18, J20, J28].

RESEARCH ASSISTANTS

- 2025 - PRESENT **Co-supervision** of Mr Marco Vitturini at the University of Turin as part of the *Digital transition and artificial intelligence in the social and healthcare sector in the Province of Cuneo: dissemination, best practices, and possible developments* project.
- 2025 - 2026 **Co-supervision** of Mr Giuseppe Martino at the University of Turin as part of the *TrustAlert* project.
- 2024 - 2025 **Co-supervision** of Ms Michela Bersia at the University of Turin as part of the *TrustAlert* project.
Co-supervision of Mr Matteo Bracco at the University of Turin as part of the *TrustAlert* project.
Co-supervision of Mr Emanuele Pietropaolo at the University of Turin as part of the *TrustAlert* project. Publications: [P6].

POSTDOCTORAL RESEARCHERS

- 2024 - 2025 **Co-supervision** of Dr Idrissa SY during his postdoc at the University of Turin. Project title: *"Extending a multilayer perceptron model to censored data"*, funded by the Ministero degli Affari Esteri e della Cooperazione Internazionale
- 2020 - 2023 **Co-supervision** of Dr Niccolò Rossi during his postdoc at King's College London (already supervised during his PhD). Publications [J31, J32, J42, J45, J46, J56, J63].
- 2016 - 2018 **Assistance with the supervision** of Dr Harriet Cullen during her research fellowship at King's College London. Publications: [J19].

INTERNS

- A.Y. 2024-25 & 25-26 **Supervisor** of Ms Triparna Day's internship as part of her MSc in Biotechnological and Chemical Sciences in Diagnostics.
- A.Y. 2018-2019 **Supervisor** of Mr Yuhao Lin's summer project as part of the King's Undergraduate Research Fellowships (KURF)
Supervisor of Ms Olivia Castellini Pérez's summer internship as part of the Erasmus+ program
- A.Y. 2017-18 **Supervisor** of Ms Lechun Huo's summer project as part of the King's Undergraduate Research Fellowships (KURF).
- A.Y. 2016-17 **Supervisor** of Ms Yuri Nemoto's summer project as part of the King's Undergraduate Research Fellowships (KURF).
Co-supervisor of Ms Fudi Wang's research visit at King's College London
- A.Y. 2015-16 **Co-supervisor** of Ms Esther Kok's summer internship at the CERN OpenLab.
- A.Y. 2013-14 **Co-supervisor** of Mr Marcin Świstak's internship at Imperial College London.

PUBLICATIONS

[†] indicates that the authors contributed equally to the work

[‡] indicates that the authors jointly supervised the work

✉ indicates that the author is a corresponding author

INTERNATIONAL JOURNALS

- [J65] Ometto S.[†] ✉, Chatterjee S.[†] ✉, Vergani A.M., Landini A., Sharapov S., Giacomuzzi E., **Visconti A.**, Bianchi E., Santonastaso F., Soda E.M., Cisternino F., Ieva F., Di Angelantonio E., Pirastu N., Glastonbury G.A. ✉, *Hundreds of cardiac MRI traits derived using 3D diffusion autoencoders share a common genetic architecture*, Nature Communications, 2026, <https://doi.org/10.1038/s41467-026-74575-y>
- [J64] Lin, Y., Kouraki, A., Cheetham, N.J. Louca P., Bowyer R.C.E., Pope R., Asnicar F., Zhang X., **Visconti A.**, Falchi M., Spector T.D., Segata N., Valdes A.M. ✉, Menni C. ✉, *Gut microbiome composition and function reflect socioeconomic deprivation*, npj Biofilms Microbiomes, 2026, 10.1038/s41522-026-00917-9
- [J63] Aliyev, E., Syed, N., **Visconti A.**, Aliyev T., Belkadi A., Ghorbani M., . . . , Akil A., Suhre K., Mokrab Y., Falchi M.[‡], and Fakhro K.A.[‡] ✉, *The biomedical landscape of genomic structural variation in the qatari population*. Nature Communications, 2026, doi:10.1038/s41467-025-67763-9
- [J62] De Luca S. ✉, Márquez Villacís J.J., Medoro C.M., Zanaga G., Grimaldi P., Dragoni M., **Visconti A.**, Berchialla P., and Rizzo G. ✉, *Outbreak Monitoring: Epidemic Surveillance through Anomaly Detection on Zero-Shot News Classification*, Expert Systems with Applications, 2025, doi:10.1016/j.eswa.2025.130852
- [J61] Pope R., **Visconti A.**, Zhang X., Louca P., Baleanu A.F., Lin Y., Asnicar F., Bermingham K., Wong K.E., Michelotti G.A., Wolf J., Segata N., Berry S.E., Spector T.D., Leeming E.R., Gibson R., Menni C.[‡], and Falchi M.[‡] ✉, *Faecal metabolites as a readout of habitual diet capture dietary interactions with the gut microbiome*, Nature Communications, 2025, doi:10.1038/s41467-025-66046-7
- [J60] Di Blasio M., Bami Z., Bracco M., Fava C., **Visconti A.** ✉, Baldi I.[‡], and Berchialla P.^{<‡}, *Exploring Patient Preference Information in Clinical Research and Decision Models: An Investigative Survey*, Patient Preference and Adherence, 2025, doi:10.2147/PPA.S541040
- [J59] Koumantakis, E., Remoundou, K., Colombi, N., Fava C., Roussaki I., **Visconti A.**[‡] ✉, and Berchialla P.[‡], *Deep learning models for ICU readmission prediction: a systematic review and meta-analysis*, Critical Care, 2025, doi:10.1186/s13054-025-05642-x
- [J58] Standl M.[†], Budu-Aggrey A.[†], Johnston L.J., Elias M.S., . . . , **Visconti A.**, . . . , Langan S.M.[‡], Paternoster L.[‡], and Brown S.J.[‡] ✉, *Gene–Environment Interaction Affects Risk of Atopic Eczema: Population and In Vitro Studies*, Allergy, 2025, doi:10.1111/all.16605
- [J57] Saravi S.S.S. ✉, Pugin B., Constancias F., Shabanian K., Spalinger M., Thomas A., Le Gludic S., Shabanian T., Karsai G., Colucci M., Menni C., Attaye I., Zhang X., Allemann M.S., Lee P., **Visconti A.**, Falchi M., Alimonti A., Ruschitzka F., Paneni F and Beer J.H. ✉, *Gut microbiota-dependent increase in phenylacetic acid induces endothelial cell senescence during aging*, Nature Aging, 2025, doi:10.1038/s43587-025-00864-8
- [J56] **Visconti A.**[†], Rossi N.[†], Bondt A., Hipgrave Ederveen A., Thareja G., Koeleman C.A.M., Stephan N., Halama A., Lomax-Browne H.J., Pickering M.C., Zhou X., Wuhler M.[‡], Suhre K.[‡], and Falchi M.[‡] ✉, *The genetics and epidemiology of N- and O- Immunoglobulin A glycomics*, Genome Medicine, 2024, doi:10.1186/s13073-024-01369-6
- [J55] Valdes A.M.[†] ✉, Louca P.[†], **Visconti A.**, Asnicar F., Bermingham K., Nogal A., Wong K., Wolf J., Michelotti G., Segata N., Spector T.D., Berry S.E., Falchi M., Menni C. ✉, *Vitamin A carotenoids, but not retinoids, mediate the impact of a healthy diet on gut microbial diversity*, BMC Medicine, 2024, doi:10.1186/s12916-024-03543-4
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